

Noel J. Young

Senior Software Engineer – Backend (Go · PHP · Python)

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Professional Summary

Software engineer with 15 years' experience, starting in marketing agency web development before moving into backend systems for telecommunications, logistics and, since 2023, e-commerce payments. Most recently a core engineer on the Nintendo of Europe payment gateway integration on Actindo's commerce platform: card, PayPal, Apple Pay and Google Pay through Nintendo's Adyen-backed payment gateway across multiple European markets, from API integration and workflow design through go-live and ongoing production support.

Works across PHP, Go and Python. Currently designing and building two SaaS products end to end: an AI-driven WhatsApp booking platform in Go (tool-calling LLM agent, RAG, EU AI Act-aware design) and a goal and habit-tracking app in Python/FastAPI.

Former hands-on engineering manager, comfortable owning a problem from the query plan to the go-live checklist and explaining the trade-offs to the business.

Professional Experience

Senior Software Engineer (Contract) – Actindo AG, Germany (remote)

Jul 2023 – Dec 2025 · Apr 2026 – Present

Contract engineer on Actindo's cloud commerce and order-management platform, implementing various integrations for enterprise clients.

Nintendo of Europe Payment Gateway (NPGW)

- Designed and implemented the API integration between Actindo and the Nintendo Payment Gateway (NPGW, backed by Adyen), covering the full payment lifecycle: card, PayPal, Apple Pay and Google Pay across various EU markets (e.g. DE, UK, FR, IT, CH) and multiple currencies.
- Built the PHP payment driver plugins on Actindo's Core1 platform: API clients, request handlers, bootstrap configuration and ETL mappings.
- Designed, built and maintain multiple Petri net-based workflows covering order-to-delivery, authorisation, capture, refunds, chargebacks and cancellations.

- Implemented complex payment scenarios: pre-authorisation, Merchant Initiated Transactions for pre-orders and backorders, partial captures, multi-currency chargebacks with mid-rate conversion, and credit note generation.
- Built error handling and retry logic for authorisation failures, including PayPal token retries and configurable error-code classification.
- Took the integration through its February 2025 production go-live, followed by ongoing production hardening, incident resolution and workflow optimisation.
- Built a Go mock server for the NPGW API, shared by all isolated sandboxes, with header-switchable scenarios (declines, 3DS, gateway errors, timeouts, rate limits). Now being integrated into the automated E2E test suite.
- Built SQL tooling to identify and resolve errored workflow processes in bulk across payment and order tables.
- Took part in onsite workshops and regular technical alignment sessions with Nintendo of Europe; wrote the workflow, testing, deployment and chargeback documentation.

Platform and performance

- Diagnosed a production search performance problem in MySQL FULLTEXT: EXPLAIN ANALYZE showed entity-type filtering applied only after a full-scope FULLTEXT scan, with no typo tolerance. Evaluated MySQL-native, PostgreSQL FTS, Elasticsearch and Meilisearch options against team skills and infrastructure.
- Separated two conflated Elasticsearch initiatives (a broken legacy CDC/Debezium pipeline and an unbuilt BI outbox consumer), giving each a clear scope for planning.
- Built the BMW Group payment driver API integration; fixed RMA (returns) workflow issues in validation, return-position linking and blocked-storage handling.

Stack: PHP, Go, MySQL, Actindo Core1, NPGW (Adyen), PayPal, Salesforce Commerce Cloud, SAP, Postman

Products (Founder & Engineer)

Botico – AI WhatsApp Booking Platform

May 2026 – Present · in development

Multi-tenant WhatsApp automation platform for appointment-based businesses (salons, clinics, gyms). Designed and built solo.

- Built as a modular Go monolith on PostgreSQL, with background workers handling reminders, retries, review requests and knowledge-base embeddings.

- Built an AI gateway that routes each message to one of four lanes: guided flow, human hand-off, cached answer, or a single tool-using LLM call, with pure-function pre-routing and guard checks before anything is sent.
- Built a tool-calling framework (registry and executor) with JSON-schema validation of tool inputs and model replies, plus versioned, per-tenant prompts.
- Built a RAG knowledge base using Voyage embeddings and pgvector (HNSW) on PostgreSQL 16, exposed to the agent as a search tool.
- Designed for EU AI Act compliance: AI disclosure, human hand-off, and an audit trail of every model call, tool call and routing decision, with per-message cost reporting.
- Multi-tenant foundations: JWT auth, per-customer advisory locking for message ordering, signature-verified Meta webhooks.
- Plugin-based integrations with Calendly, Fresha, Google Calendar and Business Profile, Stripe, SumUp and Tally.
- React and TypeScript frontend with a visual workflow builder (React Flow) and a live WebSocket dashboard.

Stack: Go, chi, PostgreSQL, pgvector, Anthropic API, Voyage AI, React, TypeScript, Vite, Tailwind, Docker

Loonya – Goal and Habit-Tracking SaaS

Aug 2026 – Present · in development

Subscription productivity app combining calendar, habits, goals, journal and gamification. Designed and built solo.

- FastAPI (Python 3.12) backend on PostgreSQL with raw SQL and no ORM, plus a React frontend; in-process background workers for reminders, search indexing and metrics.
- Hybrid semantic and fuzzy search using Meilisearch and Voyage AI embeddings. Writes flow through a transactional outbox to an async indexer, with per-user isolation enforced on a single read path.
- Stripe subscriptions (monthly and yearly, 14-day trial), webhooks, customer portal and a voucher system.
- MCP server with API-key auth that lets AI assistants read a user's own data.
- Encrypted nightly backups (pg_dump with age) with off-site copies and an automated weekly restore test; health checks, alerting, and admin trend and cohort dashboards.
- Rate limiting, CSP security headers, email verification, and full account deletion across Stripe, Google, the database, uploads and the search index.
- Around 400 backend tests and written documentation for operations, billing and infrastructure.

Stack: Python, FastAPI, PostgreSQL, Meilisearch, Voyage AI, React, Vite, Tailwind, Stripe, Docker

Earlier Experience

Software Engineer – Devprox, South Africa

Jan 2023 – Jul 2023

- Added Redis caching for stock-availability checks.
- Built a PHPUnit and cURL-based testing framework for front-end application behaviour.
- Contributed to a new finance model for Middle East clients, and to a cascading pricing model that automated exchange-rate calculations via a REST API.

Engineering Manager (hands-on) – Megafreight Services, South Africa

Mar 2019 – Dec 2022

Led architecture and core development of the freight-forwarding systems while managing the engineering team.

- Owned the financial systems: per-shipment profit and loss, accounts payable and receivable, and management dashboards showing costs by branch and client.
- Delivered enterprise integrations: Champ Cargohub air cargo (REST, containerised Java service), C.H. Robinson sea freight EDI over AS2 (Python), customs REST APIs, NetSuite (SOAP) and automated exchange rates (Fixer).
- Led SQL performance work: query analysis, indexing strategy and window-function reporting.
- Planned the legacy-to-new platform migration with data synchronisation and business continuity.
- Built a real-time shipment status system giving leadership visibility during COVID remote working.
- Hired, onboarded and mentored engineers.

Stack: PHP, Laravel, Vue.js, MySQL, Python, Java, Docker

Engineering Manager (hands-on) – Switch Telecom, South Africa

Apr 2016 – Feb 2019

Led development and modernisation of the client portal, hosted PBX platform and billing engines.

- Refactored the Asterisk configuration engine to stabilise SIP reloads; built hosted switchboard features including IVR timeouts, ring groups and number porting workflows.
- Led the technical rollout of the 2018 South African VAT change across all billing engines.
- Built a call analytics PDF reporting engine.
- Introduced formal development processes, sprint planning and documentation standards; ran technical recruitment.

Stack: PHP, Laravel, Vue.js, Asterisk, MySQL, Linux

Founder & Developer – Nift, South Africa

Jan 2015 – Dec 2016

- Built a marketplace that aggregated new-car quotes from dealerships. Signed up 150+ dealerships and reached funding discussions with ABSA and the Barclays Africa Seeker Fund.

Co-founder & Developer – Student Notes, South Africa

Jan 2013 – Dec 2016

- Built an online marketplace for students to buy and sell study notes (Drupal/PHP).

Software Developer – SABEeX, South Africa

Dec 2014 – Mar 2016

- Built a greenfield procurement management system in PHP (CodeIgniter) that is still in use today; maintained the company's Drupal websites and office server infrastructure.

Software Developer – ERS Biometrics, South Africa

Mar 2014 – Nov 2014

- Built an offline-capable time and attendance web application in plain JavaScript for users with limited internet access.

Web Developer – Starbright Solutions (marketing agency), South Africa

May 2011 – Feb 2014

- Built marketing websites on a standardised Drupal setup that sped up launches, including custom modules for the PayU and Paygate payment gateways.

Skills

- **Languages:** PHP, Go, Python, SQL, JavaScript, TypeScript
- **Backend:** Laravel, Symfony, FastAPI, chi, REST APIs, webhooks, OAuth2, JWT, background workers
- **Data:** MySQL, MariaDB, PostgreSQL (pgvector, full-text search), Redis, Meilisearch, query optimisation
- **Payments:** Adyen, Stripe, PayPal, Apple Pay, Google Pay, chargebacks, Merchant Initiated Transactions
- **AI:** LLM tool calling, RAG, embeddings, MCP, Anthropic API
- **Frontend:** React, Vue.js, Tailwind
- **Tooling:** Docker, Git, Linux, RabbitMQ, PHPUnit, Grafana

Education & Certifications

- Certificate in IT Management – University of Cape Town (2021)
- The Coding Interview Bootcamp: Algorithms + Data Structures – Udemy (2025)
- Advanced Relational Database and SQL – Coursera (2023)
- Master Redis – From Beginner to Advanced (2023)
- Certified ScrumMaster (CSM) – Scrum Alliance (2018)
- ITIL Foundation – IT Service Management (2018)
- Self-taught in programming, data structures and system design (2011 – present)

Languages

English (native) · Afrikaans (native) · Spanish (beginner, studying daily)

Interests

Ultra-distance runner: 60+ marathons, longest race 152 km, and training towards a 100-mile finish. The same discipline and consistency carry into my work.